

AI Provider – Research & Decision

Decision

Chosen: OpenRouter as the AI gateway, routing to a free model.

OpenRouter sits in front of many AI models using OpenAI compatible API. We will use its free tier for this build and can switch to a paid if we need to.

Why we needed to decide

The shared-workspace feature calls an AI model to work through a session's context. We need one provider that the backend commits to, chosen deliberately rather than by default and cheap enough for a student project that will be used as a working concept rather than a production system.

Options Compared

Option	Pros	Cons
Claude	• Strongest context handling. • Simple, well documented SDK • The behaviour we originally designed for	• Paid per token (small at our scale but not free) • Locked to one vendor • Needs a billing setup
OpenAI (ChatGPT)	• Mature SDK • Strong Models • Widely used	• Paid • Locked to one vendor • No advantage over Anthropic
OpenRouter	• OpenAI compatible API • Swap models by changing a single line • Not locked into one vendor • Free tier removes cost for a concept build	• Free models are rate limited and lower quality • Adds a small routing layer • Free tier model availability can change

Reasoning

1. **Gateway over going direct:** OpenRouter is OpenAI API compatible, so the backend talks to one interface and can target any model. That allows us to not be locked into any one vendor, if a model is retired, gets too expensive or underperforms, we can swap the model id and nothing else changes. Our code already isolates the AI call behind a single function, so this costs us almost nothing to adopt.
2. **Free model over paid:** Our project proves an architecture, that one person's context reaches another person's session automatically. That plumbing works regardless of how eloquent the model is, so a free model demonstrates the concept just as well as a paid one, at no cost. Response quality isn't what is being validated.

Trade-offs we accepted

- Free-tier models are rate-limited, heavy test runs may hit limits (we retry rather than treat a rate-limit as a bug).
- Free models are lower quality than Clade/GPT, acceptable because we are demonstrating the concept not the answer quality.
- Free-model availability on OpenRouter can change, the model is a single constant, easily swapped.

Implication for the client

Worth noting to NBN: the workspace is model agnostic. It currently runs on a free model to validate the concept, moving to a stronger paid model later is a one-line change, not a full rebuild. This is recorded in the requirements doc's Ethics & Limitations section.